

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Mid-Semester Examination, Spring 2022-23

B Tech 6th Semester

Subject Name: Computer Networks

Maximum Marks: 30

Subject Code: CS3002

Exam Duration: 2 Hours

Number of pages: 2

Note: There are **Five** questions. Attempt **All** the questions. All parts of a same question should be attempted in a sequence.

Q.1 (a) Write four differences between *Non-persistent* and *persistent* HTTP.

(b) Differentiate between *packet switching* and *circuit switching*.

(c) List out five network topologies. Which topology is fully connected? Explain with example.

(d) Explain the header information such as PSH, URG, RST, SYN, FIN, ACK, and Urgent Data Pointer etc. of TCP.

(e) Why web-caching is needed? List out the alternative of web-caching.

[2+2+2+2+2=10 Marks]

Q.2 Hosts A and B are each connected to a switch S via 100-Mbps link. The propagation delay on each link is 120 μ s. S is a store-and-forward device; it begins retransmitting a received packet 35 μ s after it has finished receiving it. Calculate the total time required to transmit 10,000 bits from A to B.

(i) As a single packet.

(ii) Assume the 4 switches that implement “cut-through” switching; it is able to begin retransmitting the packet after the first 600 bits have been received.

[2+3=5 Marks]

Q.3 (a) SMTP protocol is used for which purpose? When the mail server sends mail to other mail servers it acts as a client or a server? Explain your reasoning.

(b) Consider a NITR network having capacity of 25Mbps connected to internet through 10 Mbps access link. Consider average object size is 2,500,000 bits, average request rate from institution’s browsers to origin servers is 2 requests/sec and delay from institutional access router to any origin server and back to router = 4 sec. Calculate the following

(i) Traffic intensity on the LAN and traffic intensity on the access link.

(ii) Total response time at host of NITR.

[2+3=5 Marks]

Q.4 (a) What do you mean by *congestion* in a network? How it affects the *Throughput* when retransmission occurs due to loss and duplicate segments are transmitted? Plot the graph and explain your reasoning.

(b) Consider an instance of TCP’s Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2MSS and the threshold at the start of the first transmission is 8MSS. Assume that a time-out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

[2+3=5 Marks]

Q.5 UDP and TCP use 1's complement for their checksum. Suppose you have the following three 8-bit bytes: 01010011, 01100110, 01110100.

- (i) What is the 1's complement of the sum of these 8-bit bytes? (Note that although UDP and TCP use 16-bit words in computing the checksum, for this problem you are being asked to consider 8-bit sum.) Show all work.
- (ii) Why is it that UDP takes the 1's complement of the sum: that is, why not just use the sum?
- (iii) With the 1's complement scheme, how does the receiver detect errors?
- (iv) Is it possible that a 1-bit error will go undetected?
- (v) How about a 2-bit error?

[1+1+1+1+1=5 Marks]